

CSSS Math Camp 2026

September 14 -18, 2026

Website: <https://www.csss.washington.edu/academics/math-camp>

Canvas: See Canvas (<https://canvas.uw.edu>) for course materials once registered

Lecture Location: PAR 120

Lab Location: PAR 160

Time: 9am – 4pm

Curriculum Overview:

Day 1:

- Algebra, functions, matrix algebra
- Introduction to R/RStudio, Quarto, project management

Day 2:

- Basic calculus - derivatives and integrals
- Data manipulation in base R and tidyverse

Day 3:

- Probability, random variables, distributions
- Data manipulation and visualization

Day 4:

- Probability and distributions, expected value and variance
- Probability and simulation in R

Day 5:

- Hypothesis testing, inference, confidence intervals
- Exploratory data analysis in R

Approximate Daily Schedule:

9:00-10:30	Lecture on math concepts for the day
10:30-10:45	Break
10:45-12:00	Lecture part 2 and practice problems
12:00 - 1:15	Lunch
1:15-2:30	Guided R lab
2:30-2:45	Break
2:45-4:00	Working session for R practice problems and class discussion